

Week 8: Bootstrap

MATH-517 Statistical Computation and Visualization

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2025-11-07

Introduction

- population F
- random sample $\mathcal{X} = \{X_1, \dots, X_N\}$ from F
- characteristic of interest $\theta = \theta(F)$

Goal: Extract information about θ using $\mathcal{X}$ and find reliable frequentist assessment of uncertainty

Leading Example: The mean $\theta = \mathbb{E}(X_1) = \int x dF(x)$

Δ

F can be estimated:

- parametrically
 - assuming $F \in \{F_\lambda \mid \lambda \in \Lambda \subset \mathbb{R}^p\}$ for some integer p , take $\widehat{F} = F_{\widehat{\lambda}}$ for an estimator $\widehat{\lambda}$ of the parameter vector λ obtained by, e.g., MLE
- non-parametrically
 - by the ECDF, i.e., $\widehat{F} = \widehat{F}_N$ where $\widehat{F}_N(x) = \frac{1}{N} \sum_{n=1}^N \mathbb{1}_{[X_n \leq x]}$

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- characteristic of interest $\theta = \theta(F)$

Leading Example: The mean $\theta = \mathbb{E}X_1 = \int x dF(x)$

- parametrically: $\hat{\theta} = \int x dF_{\hat{\lambda}}(x)$
- non-parametrically: $\hat{\theta} = \int x d\hat{F}_N(x) = \frac{1}{N} \sum_{n=1}^N X_n$

Δ

Key questions

- How does $\hat{\theta}$ behave when samples are repeatedly taken from F ?
- How can we use knowledge of this to learn about θ ?

Introduction: Thought Experiment

Imagine F is known. Then, we could answer the questions by

- analytical calculation
- Monte Carlo simulation

For $r = 1, \dots, R$:

- generate random sample $x_1^*, \dots, x_N^* \stackrel{\text{i.i.d.}}{\sim} F$
- compute $\hat{\theta}_r^*$ using $x_1^*, \dots, x_N^*$
- output after R iterations:

$$\hat{\theta}_1^*, \hat{\theta}_2^*, \dots, \hat{\theta}_R^*$$

Use $\hat{\theta}_1^*, \hat{\theta}_2^*, \dots, \hat{\theta}_R^*$ to estimate **sampling distribution** of $\hat{\theta}$

$\Rightarrow$ If $R \rightarrow \infty$, then get perfect match to theoretical calculation (if available), i.e., Monte Carlo error disappears completely. In practice R is finite, so some error remains

Introduction

- population F
- random sample $\mathcal{X} = \{X_1, \dots, X_N\}$ from F
- characteristic of interest $\theta = \theta(F)$ (emphasize dep. on F)
- sample characteristic $\hat{\theta} = \theta(\hat{F})$
- **sampling distribution** of $\hat{\theta}$
 - bias or MSE needed to rate the estimator - all characteristics of sampling distribution
 - quantiles of sampling distribution needed for CIs or testing on θ

Leading Example: The mean $\theta = \mathbb{E}(X_1) = \int x dF(x)$

- non-parametrically: $\hat{\theta} = \int x d\hat{F}_N(x) = \frac{1}{N} \sum_{n=1}^N X_n$
- if F is Gaussian, then $\hat{\theta} \sim \mathcal{N}(\theta, \frac{\sigma^2}{N})$ is the sampling distribution
 - without Gaussianity, there is still a sampling distribution, we just don't know what it is (most of the time)

Δ

Introduction

Inference about θ is based on the **sampling distribution**, which is given by the sampling process

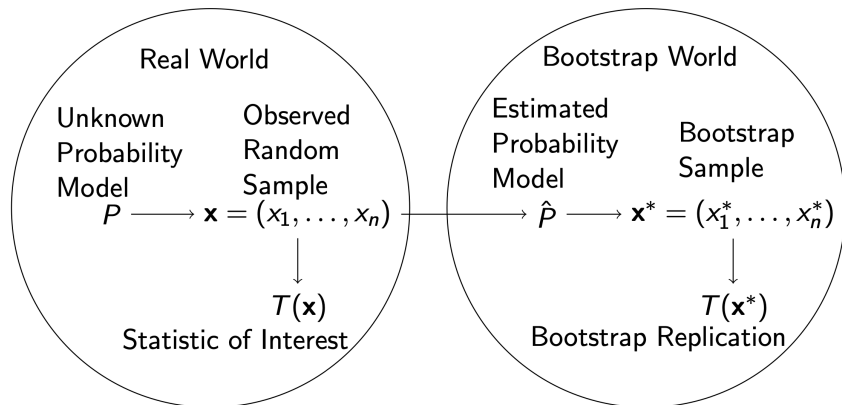
- If we control the sampling process, we can approximate the sampling distribution by Monte Carlo
- F unknown but $\widehat{F}$ is known. Then, the (re)sampling distribution can be studied/approximated by Monte Carlo

The Bootstrap Idea: The (re)sampling process from $\widehat{F}$ can mimic the sampling process from F itself

Sampling (real world): $F \Rightarrow X_1, \dots, X_N \Rightarrow \hat{\theta} = \theta(\widehat{F})$

Resampling (bootstrap world): $\widehat{F} \Rightarrow X_1^*, \dots, X_N^* \Rightarrow \hat{\theta}^* = \theta(\widehat{F}^*)$

Illustration



⇒ removes need for mathematical skills but still perform well in practice (usually!)

Principle of the Non-Parametric Bootstrap

Bootstrapping an estimator $\hat{\theta} = g(X_1, \dots, X_N)$ can be done as follows

- Generate a **bootstrap sample**

$$X_1^*, \dots, X_N^* \stackrel{\text{i.i.d.}}{\sim} \hat{F}_N$$

(take N uniform random draws with replacement from the original dataset $\{X_1, \dots, X_N\} \Rightarrow$ **resampling the data**)

- Compute the bootstrapped estimator

$$\hat{\theta}^* = g(X_1^*, \dots, X_N^*)$$

- Repeat the first two steps B times to obtain $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$

As $N \rightarrow \infty$ and $B \rightarrow \infty$, bootstrap sample moments of $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$ converge to the corresp. sample moments of sampling distribution of $\hat{\theta}$

Question: What about the parametric bootstrap?

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Question: What about the parametric bootstrap? replace $\hat{F}_N$ by a parametric estimate $\hat{F}$

Using the $\hat{\theta}^{\star b}$ to estimate Standard Errors

Bootstrap replicates $\hat{\theta}^{\star b}$ used to assess quality of $\hat{\theta}$

- Variance of $\hat{\theta}$ as estimator of θ is

$$\begin{aligned}\text{Var}(\hat{\theta}) &= \mathbb{E}_F[\{\hat{\theta} - \mathbb{E}_F(\hat{\theta})\}^2] \\ &= \mathbb{E}[\{\hat{\theta} - \mathbb{E}(\hat{\theta} \mid X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} F)\}^2 \mid X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} F)]\end{aligned}$$

Moving from the real world to the bootstrap world,

$$\begin{aligned}\widehat{\text{Var}}(\hat{\theta}) &= \mathbb{E}[\{\hat{\theta} - \mathbb{E}(\hat{\theta} \mid X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} \hat{F})\}^2 \mid X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} \hat{F}] \\ &= \mathbb{E}[\{\hat{\theta}^{\star} - \mathbb{E}(\hat{\theta}^{\star})\}^2] \\ &\approx \frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{\star b} - \bar{\hat{\theta}}^{\star})^2,\end{aligned}$$

i.e., the sample variance of the bootstrap replicates estimates the variance of the estimator (real world)

Using the $\hat{\theta}^{\star b}$ to estimate the Bias

Bootstrap replicates $\hat{\theta}^{\star b}$ used to estimate properties of $\hat{\theta}$

- Bias of $\hat{\theta}$ as estimator of θ is

$$\text{bias}(\hat{\theta}) = \text{bias}(F) = \mathbb{E}(\hat{\theta} \mid X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} F) - \theta(F)$$

estimated by replacing unknown F by known estimate $\hat{F}$

$$\begin{aligned}\text{bias}(\hat{F}) &= \mathbb{E}(\hat{\theta} \mid X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} \hat{F}) - \theta(\hat{F}) \\ &= \mathbb{E}(\hat{\theta}^{\star}) - \hat{\theta}\end{aligned}$$

- Replace theoretical expectation by empirical average

$$\widehat{\text{bias}}(\hat{\theta}) = \text{bias}(\hat{F}) \approx \bar{\hat{\theta}}^{\star} - \hat{\theta} = B^{-1} \sum_{b=1}^B \hat{\theta}^{\star b} - \hat{\theta}$$

Question: How can we use this to improve inference?

Bias Correction: Another Example

- $X_1, \dots, X_N$ i.i.d. with $\mathbb{E}|X_1|^3 < \infty$
- characteristic of interest: $\theta = \mu^3$, where $\mu = \mathbb{E}(X_1)$
- empirical estimator: $\hat{\theta} = (\int x d\hat{F}_N)^3 = (\bar{X}_N)^3$ is biased
 - bias $b := \text{bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta$ of order N^{-1}
- bootstrap: estimate the bias b as $\hat{b}^*$
- bias-corrected estimator

$$\hat{\theta}_b^* = \hat{\theta} - \hat{b}^*$$

has smaller order bias (order N^{-2})

Something similar happens more generally for $\theta = g(\mu)$ when g is sufficiently smooth

Bias Correction: Another Example

- $X_1, \dots, X_N$ i.i.d. with $\mathbb{E}|X_1|^3 < \infty$
- Interest in $\theta = \mu^3$, where $\mu = \mathbb{E}(X_1)$, $\sigma^2 = \mathbb{E}(X_1 - \mu)^2$, and $\gamma = \mathbb{E}(X_1 - \mu)^3$
- estimator: $\hat{\theta} = \left(\int x d\hat{F}_N\right)^3 = (\bar{X}_N)^3$ is biased

$$\mathbb{E}_F(\hat{\theta}) = \mathbb{E}_F(\bar{X}_N^3) = \mathbb{E}\left[\mu + N^{-1} \sum_{n=1}^N (X_n - \mu)\right]^3 = \mu^3 + \underbrace{N^{-1}3\mu\sigma^2 + N^{-2}\gamma}_{=b=\mathcal{O}(N^{-1})}$$

- bootstrap: estimate the bias $b := \text{bias}(\hat{\theta}) = \mathbb{E}\hat{\theta} - \theta$ as $\hat{b}^*$

$$\begin{aligned}\mathbb{E}_{\hat{F}_N}\hat{\theta}^* &= \mathbb{E}_{\hat{F}_N}\{(\bar{X}_N^*)^3\} = \mathbb{E}_{\hat{F}_N}\left\{\bar{X}_N + N^{-1} \sum_{n=1}^N (X_n^* - \bar{X}_N)\right\}^3 \\ &= \bar{X}_N^3 + \underbrace{N^{-1}3\bar{X}_N\hat{\sigma}^2 + N^{-2}\hat{\gamma}}_{=\hat{b}^*}\end{aligned}$$

- bias-corrected estimator: $\hat{\theta}_b^* = \hat{\theta} - \hat{b}^*$ has smaller order bias

$$\mathbb{E}_F(\hat{\theta}_b^*) = \mu^3 + N^{-1}3 \underbrace{\{\mu\sigma^2 - \mathbb{E}_F(\bar{X}_N\hat{\sigma}^2)\}}_{\mathcal{O}(N^{-1})} + N^{-2} \underbrace{\{\gamma - \mathbb{E}_F(\hat{\gamma})\}}_{\mathcal{O}(N^{-1})}$$

Leading Example: Using the $\hat{\theta}^{*b}$ for CI

- $X_1, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} F$ and $\theta = \theta(F) = \int x dF$
- $\hat{\theta} = \bar{X}_N$ and $\hat{\sigma}^2 = (N-1)^{-1} \sum_{n=1}^N (X_i - \bar{X}_N)^2$
- we want θ_α such that $P\{\theta \geq \theta_\alpha\} = 1 - \alpha$, for $0 < \alpha < 1$

1 Exact CI. (rare) Assuming Gaussianity,

$$T = \sqrt{N} \frac{\bar{X}_N - \theta}{\hat{\sigma}} \sim t_{N-1} \quad \Rightarrow \quad P\{T \leq t_{N-1}(1 - \alpha)\} = 1 - \alpha$$

and so we get a CI with exact coverage

$$\theta \geq \bar{X}_N - \frac{\hat{\sigma}}{\sqrt{N}} t_{N-1}(1 - \alpha) := \hat{\theta}_\alpha$$

2 Asymptotic CI. Assuming only $\mathbb{E}X_1^2 < \infty$, $T \xrightarrow{d} \mathcal{N}(0, 1)$ and thus

$$P\{\theta \geq \hat{\theta}_\alpha\} \approx 1 - \alpha \quad \text{for} \quad \hat{\theta}_\alpha = \bar{X}_N - \frac{\hat{\sigma}}{\sqrt{N}} z(1 - \alpha)$$

Leading Example: Using the $\hat{\theta}^{*b}$ for CI

3 **Bootstrap CI.** Let $\mathbb{E}X_1^2 < \infty$ and $X_1^*, \dots, X_N^*$ be a bootstrap sample from the ECDF $\hat{F}_N$

- get $\bar{X}_N^* = N^{-1} \sum_{n=1}^N X_n^*$ and $\hat{\sigma}^{*2} = \frac{1}{N-1} \sum_{n=1}^N (X_n^* - \bar{X}_N^*)^2$
- set up the bootstrap statistic $T^* = \sqrt{N} \frac{\bar{X}_N^* - \bar{X}_N}{\hat{\sigma}^*}$
- B bootstrap copies $T_1^*, \dots, T_B^*$ used to estimate the dist. of T

Data		Resamples
$x = \{X_1, \dots, X_N\}$	$\Rightarrow$	$\begin{cases} \mathcal{X}_1^* = \{X_{1,1}^*, \dots, X_{1,N}^*\} & \Rightarrow T_1^* \\ \vdots & \vdots \\ \mathcal{X}_B^* = \{X_{B,1}^*, \dots, X_{B,N}^*\} & \Rightarrow T_B^* \end{cases}$

- take $q^*(1 - \alpha)$ the sample $(1 - \alpha)$ -quantile of $T_1^*, \dots, T_B^*$
- instead of $\hat{\theta}_\alpha = \bar{X}_N - \frac{\hat{\sigma}}{\sqrt{N}} z(1 - \alpha)$, consider

$$\hat{\theta}_\alpha^* = \bar{X}_N - \frac{\hat{\sigma}}{\sqrt{N}} q^*(1 - \alpha)$$

Leading Example: Coverage Comparison

2 **Asymptotic CI.** $T = \sqrt{N} \frac{\bar{X}_N - \theta}{\hat{\sigma}} \sim \mathcal{N}(0, 1)$

By the Berry-Esseen theorem

$$\begin{aligned} P_F(T \leq x) - \Phi(x) &= \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad \text{for all } x \\ \Rightarrow P\left(\theta \geq \underbrace{\bar{X}_N - \frac{\hat{\sigma}}{\sqrt{N}} z(1-\alpha)}_{=\hat{\theta}_\alpha}\right) &= P\{T \leq z(1-\alpha)\} \\ &= 1 - \alpha + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \end{aligned}$$

I.e., the coverage of the asymptotic CI is exact up to $\mathcal{O}(N^{-1/2})$

Leading Example: Coverage Comparison

3 Bootstrap CI. (assuming “ideal” bootstrap with infinite nbr of replicates)

From Edgeworth expansions (complicated!):

$$P_F(T \leq x) = \Phi(x) + \frac{1}{\sqrt{N}}a(x)\phi(x) + \mathcal{O}\left(\frac{1}{N}\right)$$

$$P_{\hat{F}_N}(T^* \leq x) = \Phi(x) + \frac{1}{\sqrt{N}}\hat{a}(x)\phi(x) + \mathcal{O}\left(\frac{1}{N}\right)$$

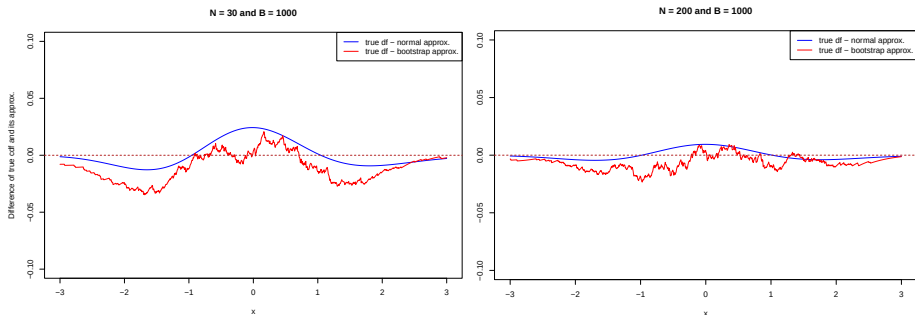
where $\hat{a}(x) - a(x) = \mathcal{O}(N^{-1/2})$

Hence, $P_F(T \leq x) - P_{\hat{F}_N}(T^* \leq x) = \mathcal{O}\left(\frac{1}{N}\right)$ and

$$\begin{aligned} \Rightarrow P\left(\theta \geq \underbrace{\bar{X}_N - \frac{\hat{\sigma}}{\sqrt{N}}q^*(1-\alpha)}_{=\hat{\theta}_\alpha^*}\right) &= P_F\{T^* \leq q^*(1-\alpha)\} + \mathcal{O}\left(\frac{1}{N}\right) \\ &= 1 - \alpha + \mathcal{O}\left(\frac{1}{N}\right) \end{aligned}$$

I.e. the coverage of the bootstrap CI is exact up to $\mathcal{O}(N^{-1})$: faster conv. rate

Leading Example: Sampling Distribution of T



Problem (1) with the non-parametric bootstrap

Use non-parametric bootstrap to estimate characteristics of the **median**

For a sample of size $N = 2m + 1$, possible distinct values of $\hat{\theta}^*$ are $X_{(1)} < \dots < X_{(N)}$, and

$$P(\hat{\theta}^* > X_{(l)}) = \sum_{r=0}^m \binom{N}{r} \left(\frac{l}{N}\right)^r \left(1 - \frac{l}{N}\right)^{N-r}$$

- exact calculations of mean, variance (etc.) of bootstrap distribution are possible and converge to correct values (as $N \rightarrow \infty$)

$\Rightarrow$ consistency holds

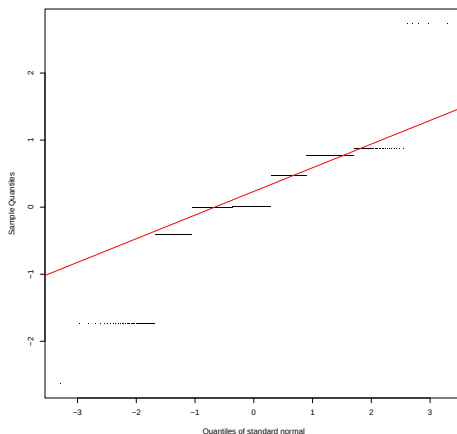
- but $\hat{\theta}^*$ concentrated on sample values and very vulnerable to unusual values

$\Rightarrow$ discreteness makes convergence very slow

E.g., bootstrap variance of the median can be very poor for heavy-tailed distributions and small sample sizes

Problem (1) with the non-parametric bootstrap

- Simulate from a sample with $N = 11$ from (standard) Cauchy
- Compute medians from $B = 1000$ bootstrap samples and center with true median ($=0$)



Problem (2) with the non-parametric bootstrap

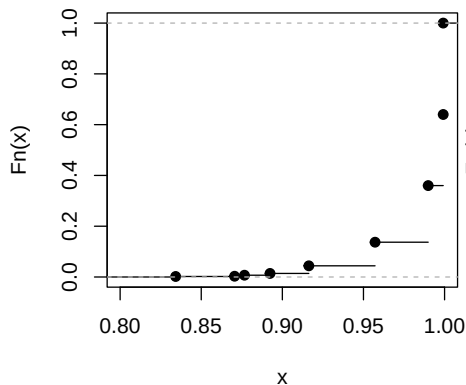
- $X_1, \dots, X_N \sim U(0, \theta)$ i.i.d., $\theta > 0$
- MLE: $\hat{\theta} = \max(X_1, \dots, X_N)$
 - $T = N(\theta - \hat{\theta})/\theta \sim \text{Exp}(1)$
- Non-parametric bootstrap: $X_1^*, \dots, X_N^*$ sampled indep. from $X_1, \dots, X_N$ with replacement
- Bootstrap estimate $\hat{\theta}^* = \max(X_1^*, \dots, X_N^*)$
 - $T^* = N(\hat{\theta} - \hat{\theta}^*)/\hat{\theta}$
- Large probability mass at $\hat{\theta}$. In fact
$$P(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/N)^N \xrightarrow{N \rightarrow \infty} 1 - e^{-1} \approx .632$$

$\Rightarrow$ the limiting distribution of T^* cannot be $\text{Exp}(1)$

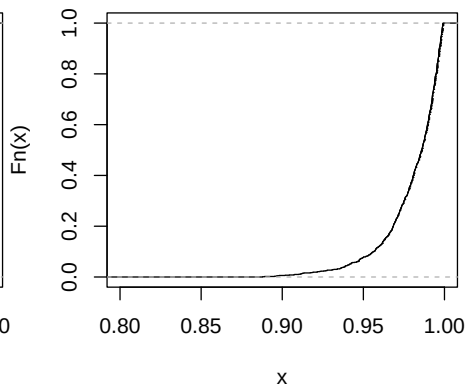
Bootstrap fails here and we will see why (consistency fails!)

Problem (2) with the non-parametric bootstrap

Nonparametric bootstrap



Parametric bootstrap



(Non-parametric) Bootstrap: Summary

- let $\mathcal{X} = \{X_1, \dots, X_N\}$ be a random sample from F
- quantity of interest: $\theta = \theta(F)$
- (plug-in) estimator: $\hat{\theta} = \theta(\hat{F}_N)$
 - write $\hat{\theta} = \theta[\mathcal{X}]$, since $\hat{F}_N$ and thus the estimator depends on the sample
- the distribution $F_{T,N}$ of a scaled estimator $T = g(\hat{\theta}, \theta) = g(\theta[\mathcal{X}], \theta)$ is of interest, e.g., $T = \sqrt{N}(\hat{\theta} - \theta)$

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The workflow of the bootstrap is as follows for some $B \in \mathbb{N}$:

	Data		Resamples
$\mathcal{X} = \{X_1, \dots, X_N\}$	$\Rightarrow$	$\left\{ \begin{array}{l} \vdots \\ \vdots \end{array} \right.$	$X_1^* = \{X_{1,1}^*, \dots, X_{1,N}^*\} \Rightarrow T_1^* = g(\theta[\mathcal{X}_1^*], \theta[\mathcal{X}])$
			$\vdots$
			$X_B^* = \{X_{B,1}^*, \dots, X_{B,N}^*\} \Rightarrow T_B^* = g(\theta[\mathcal{X}_B^*], \theta[\mathcal{X}])$

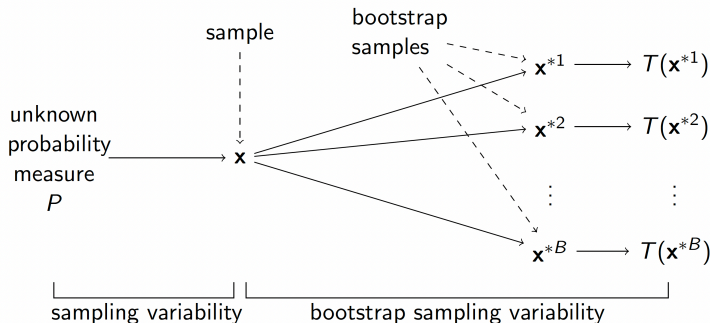
$F_{T,N}$ now estimated by $\hat{F}_{T,B}^*(x) = B^{-1} \sum_{b=1}^B \mathbb{I}_{[T_b^* \leq x]}$

- any characteristic of $F_{T,N}$ can be estimated by the char. of $\hat{F}_{T,B}^*(x)$

Bootstrap: Summary

Bootstrap combines

- the plug-in principle: sample is used to estimate $F (\approx \hat{F})$
- Monte Carlo principle: simulation replaces theoretical calculation
- two sources of variability
 - sampling variability (we only have a sample of size N)
 - bootstrap resampling variability (only B bootstrap samples)



Bootstrap: Common Questions

- How many bootstraps/Monte Carlo draws?
 - $B \geq 200$ to estimate bias or variance (next week)
 - $B = 10^3$ is taken most commonly
 - $B \geq 10^4$ better for small/large quantiles

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- Why take resamples of size N ?
 - to mimic sampling properties of samples like the original one
 - sometimes we take $m < N$ to achieve validity of bootstrap, e.g., for extreme quantiles or median (to avoid discreteness)

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- Why resample from the EDF?
 - Non-parametric MLE of F , so it's natural when no restrictions on F
 - Smooth estimate of the EDF (KDE) can be used when discreteness is severe, e.g. the case of the median

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- When does the bootstrap work (“work” = consistency)?

Consistency

Bootstrap setup:

- $T = g(X_1, \dots, X_N \mid F)$ is a scaled estimator with unknown (wanted) distribution $F_{T,N}$, with $g(X_1, \dots, X_N \mid \cdot)$ continuous
- bootstrap statistic $T^* = g(X_1^*, \dots, X_N^* \mid \hat{F})$ has $F_{T,N}^*$ also unknown
- the Monte Carlo proxy $\hat{F}_{T,B}^*$ is used instead of $F_{T,N}^*$

Glivenko-Cantelli:

$$\sup_x \left| \hat{F}_{T,B}^*(x) - F_{T,N}^*(x) \right| \xrightarrow{a.s.} 0 \quad \text{as } B \rightarrow \infty$$

Question: Under which conditions the bootstrap “works” (gives mathematically correct answers), i.e.,

$$F_{T,N}^* \rightarrow F_{T,N}, \quad \text{as } N \rightarrow \infty$$

Consistency

- ① $F_{T,N}$ must converge weakly to some continuous limit $F_{T,\infty}$

$$\int h(t) dF_{T,N}(t) \rightarrow \int h(t) dF_{T,\infty}(t) \quad \text{as } n \rightarrow \infty \text{ and } \forall h \text{ integrable}$$

$\Rightarrow$ to ensure that the wanted distribution converges to a non-degenerate limit

- ② the convergence must be uniform

$\Rightarrow$ to ensure that $F_{T,N}^*$ approaches $F_{T,\infty}$ for all possible sequences of $\hat{F}$ (which changes as N increases)

Then, the bootstrap is consistent, i.e., $\forall t$ and $\epsilon > 0$

$$P\{|F_{T,N}^*(t) - F_{T,\infty}(t)| > \epsilon\} \xrightarrow{n \rightarrow \infty} 0$$

Remark: second condition fails in the case of the maximum of a uniform!

Consistency for Smooth Transformation of the Mean

Conditions that ensure consistency of the bootstrap are guaranteed for smooth transformations of the sample mean

Theorem: Let $X_1, \dots, X_N$ be i.i.d. s.t. $\mathbb{E}(X_1^2) < \infty$ and $T = h(\bar{X}_N)$, where h is continuously differentiable at $\mu = \mathbb{E}(X_1)$ and such that $h(\mu) \neq 0$. Then

$$\sup_x \left| F_{T,N}^*(x) - F_{T,N}(x) \right| \xrightarrow{a.s.} 0 \quad \text{as } N \rightarrow \infty$$

Remarks

- bootstrap should not be used blindly
 - verification via theory
 - and/or via simulations
- folk knowledge
 - typically “works” when T asymptotically normal and data i.i.d.
 - “doesn’t work” when working with
 - statistics that do not exist (mean of Cauchy distribution)
 - non-smooth transformations of the sample (sample quantiles): non-parametric bootstrap still valid but may not work well for finite samples
 - non-i.i.d. regimes (e.g. time series): see block bootstrap or bootstrap in regression settings
- bootstrap replaces analytic calculations (in particular the Delta method), but showing that it actually works requires even deeper analytic calculations
- faster rates can be achieved by bootstrap
 - hard to prove, but often happens, e.g., when working with a skewed distribution

References

Davison & Hinkley (2009) Bootstrap Methods and their Application

Wasserman (2005) All of Nonparametric Statistics

Shao & Tu (1995) The Jackknife and Bootstrap

Hall (1992) The Bootstrap and Edgeworth Expansion

Assignment 7

Go to [Assignment 7](#) for details